

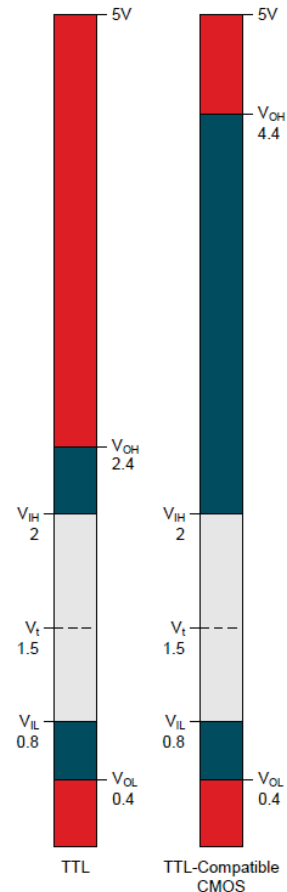
List of project tasks

1. Logic level checker with RGB LED indicator.

Although transistor-transistor-logic (TTL) digital circuits are still in use due to their robust noise immunity, applications frequently mix them with complementary metal-oxide-semiconductor (CMOS) logic circuitry. Compatibility is maintained if they are operated from 5 V supply and TTL-compatible CMOS devices (such as the HCT family) are able to output an elevated logic high voltage V_{OH} . (see **Section 5.3** of **SN74HCT00** as a reference). To ensure specification and compatibility of either devices, their output voltages V_O need to be checked for correct logic levels (according to the diagram) at a specified load current I_L .

Design a circuit that can check these levels at a maximum load current of $I_L = \pm 0.4$ mA. Indicate these levels using the **B5-4RGB-CBA** 3-color LED as follows:

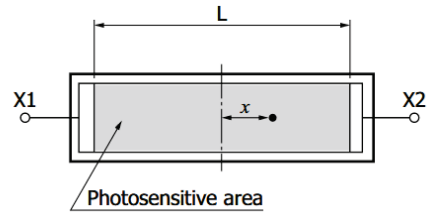
- Blue LED lights up if $V_O = V_{OL} \leq 0.4$ V.
- Green LED lights up if $V_O = V_{OH} \geq 4.4$ V.
- Red LED lights up if 0.4 V $< V_O < 2.4$ V.
- The special case, when 4.4 V $> V_{OH} \geq 2.4$ V, is to be warned by **yellow** colour. Yellow is produced, if the same current flows on both the red and green LEDs, simultaneously.



The circuit should operate from a single +5 V power supply. Drive each LED with approximately the same current. Plot the $I_L(V_O)$ characteristics of the input stage.

2. Position sensing with panel meter readout and zero indicator.

One-dimensional position sensitive detectors (PSD) are designed for precision distance measurement such as displacement meters. The **S3931** has a photosensitive area of $1 \times 6 \text{ mm}^2$ and consists of two common cathode photodiodes. Signed position of the center of a focused laser spot can be found as being proportional to the difference of the photocurrents, $x \sim I_{X2} - I_{X1}$. Locate the exact formula in the datasheet.



Design a position sensor that measures the position of a laser spot from the center and displays it using the **LDP-135-340** digital panel meter pre-configured to measure the voltage range of $0 \dots \pm 2 \text{ V}$. Assume that a red laser diode is used for which wavelength the photosensitivity is 0.4 A/W . The laser power is 0.2 mW and the beam is focused to a $200 \text{ }\mu\text{m}$ spot. Assume that the focused spot always falls entirely to the photosensitive area, in which case the maximum photocurrent is constant, $I_{\text{max}} = I_{X1} + I_{X2}$.

- Calculate I_{max} based on the given optical parameters.
- Calculate the circuit parameters so that the numerical value of the output voltage in the $V_{\text{read}} = \pm 2 \text{ V}$ range directly represent the position of the light spot in the range of $x = \pm 2 \text{ mm}$ from the PSD center.
- The special position, when the laser spot is centered so that $|x| \leq 0.1 \text{ mm}$, is to be indicated using a small power green LED of arbitrary type.
- The limiting position, when the laser spot is outside the measured range, $|x| > 2 \text{ mm}$, is to be warned using a small power red LED of arbitrary type.

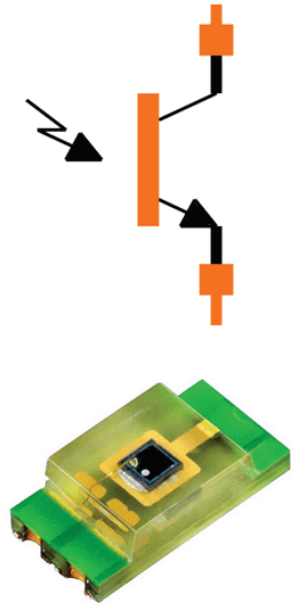
Define power supply by considering photodiode bias of $V_R = 5 \text{ V}$ as well as panel meter supply requirements. Plot the $V_{\text{read}}(x)$ curve.

3. Ambient light detector with white LED illuminator.

It is often necessary to gradually turn on illumination as ambient light decreases. Design a circuit using the **TEMT6000** phototransistor (adapted to human eye responsivity with sensitivity in the visible spectral range around 570 nm) to detect change in ambient light and gradually turn on the white light illuminator based on the **LEDSW50** medium power white LED. The LED should:

- Start lighting at $E_{v,max} = 1000 \text{ lx}$ (lux) corresponding to illuminance during cloudy daylight;
- Reach full power at $E_{v,min} = 10 \text{ lx}$ (twilight after sunset);
- Turn off above maximum illuminance, $E_v > E_{v,max}$. This state must be indicated using a small power green LED of arbitrary type.

The circuit should operate from a single power supply: specify the appropriate supply voltage. Plot the $I_{LED}(I_{sensor})$ current characteristics of the circuit with the corresponding optical power curve $P_{o,LED}(E_v)$.

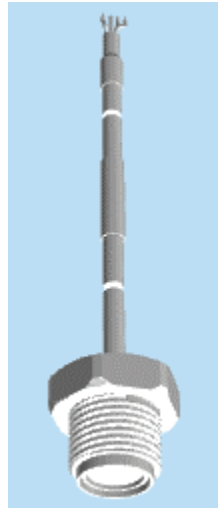


4. Water level sensor with DC motor control.

The **LM31-00000F-.20BG** is a submersible liquid level sensor that, when put into a certain depth in the liquid, measures the hydrostatic pressure of the liquid column above and converts it into a proportional voltage. The liquid height range is 0 ... 2 m (0 ... 0.2 bar) yielding 0.5 V ... 4.5 V at the sensor output. Design a circuit to measure water level in the range of $h = 0 \dots 1 \text{ m}$ (0 ... 100% for 0 ... 0.1 bar) and drive a small DC motor **PEL00884** for water pumping. The motor should:

- Rotate at its maximum speed of 2300 RPM (rotation per minute: corresponding to its maximum efficiency) when the water level is at 10%;
- Gradually slow down to minimum RPM defined at maximum current of 300 mA when the water level is at 90%;
- Turn off after reaching the maximum level of 100% to cease power consumption. Indicate this condition by using a small power green LED of arbitrary type.

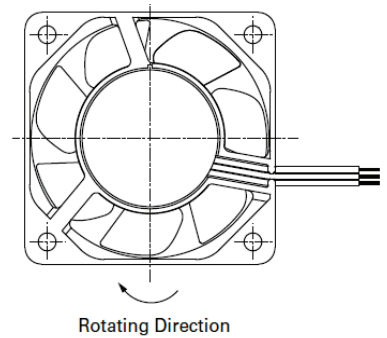
Estimate a linear $RPM(I_{\text{motor}})$ curve from the datasheet. Specify, why supply voltage stability is important, and plot the $I_{\text{motor}}(V_{\text{sensor}})$ characteristics of the circuit with the corresponding $RPM(h)$ curve.



5. Fan control with NTC thermistor.

The resistance of the negative temperature coefficient (NTC) thermistors decreases with increasing temperature. Design a circuit using the **NTCLE100E3103** thermistor with nominal resistance of $R_{25} = 10 \text{ k}\Omega$ to measure the temperature of a device housing and control a fan of type **109R0605M401** for its cooling. The fan should operate:

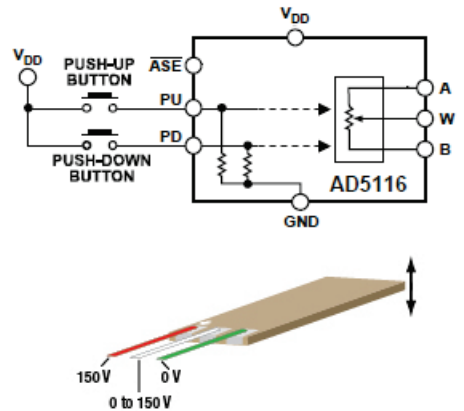
- At maximum speed with 5.5 V driving voltage if the temperature is $T \geq 50^\circ\text{C}$;
- At its rated speed with 5 V drive if the temperature is $T \geq 40^\circ\text{C}$;
- At minimum speed with 4.5 V drive if $T \geq 30^\circ\text{C}$.



In addition, utilize a small power green LED of arbitrary type to indicate that the temperature is normal, that is, the fan is off. The circuit should operate from a single power supply: specify the appropriate supply voltage. Plot the $V_{\text{fan}}(R_{\text{NTC}})$ characteristics of the circuit and the corresponding $AF(T)$ airflow characteristics (measured in $\frac{\text{m}^3}{\text{min}}$) at a maintained static pressure of 5 Pa.

6. Piezoelectric bender driver with digital potentiometer.

The **AD5116BCPZ10** is a 64 position, 10 k Ω potentiometer resistance of which can be adjusted upward or downward by two pushbutton switches. Design a circuit that employs the circuit shown (top) to generate an input voltage range of $V_{in} = 0 \dots +5 \text{ V}$ in order to change the displacement $0 \dots \pm d_{pz,max}$ of a piezo-bender **PB4NB2W** accordingly:



- At $V_{in} = 0 \text{ V}$ $d_{pz} = -d_{pz,max}$.
- At $V_{in} = +2.5 \text{ V}$ $d_{pz} = 0$.
- At $V_{in} = +5 \text{ V}$ $d_{pz} = +d_{pz,max}$.

Additionally, utilize small power blue and green LEDs of arbitrary types to indicate the bending direction (negative: blue, positive: green). For safe operation, allow only $V_{pz,max} = \pm 30 \text{ V}$ maximum piezo driving voltage: specify $\pm d_{pz,max}$. Polarity reversal is allowed with **Option 2 differential driving** of the datasheet. Specify the appropriate supply voltages by keeping in mind that the digital potentiometer requires a single +5 V, whereas the piezo driving stage requires a supply to cover the piezo driving voltage range. Plot the $V_{pz}(V_{in})$ characteristics of the circuit with the corresponding $d_{pz}(V_{in})$ curve.

7. Temperature sensor with dual colour LED indicator and voltmeter readout.

The **AD590JF** is a flat-pack 2-terminal temperature transducer that produces an output current proportional to **absolute** temperature. For supply voltages between 4 ... 30 V, the device acts as a high impedance, constant current regulator passing $\Delta I_T = 1 \frac{\mu\text{A}}{\text{K}} \Delta T$.



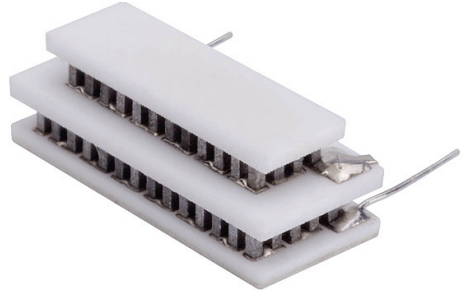
Design a circuit that measures ambient temperature in the range of $T = -20 \dots +50^\circ\text{C}$ and produces Celsius readings directly readable by a digital voltmeter as $-2 \dots +5 \text{ V}$. Moreover, apply the **TJ-L5FYTXHMNPLC2R6B** dual colour LED to indicate temperatures in discomfort zones:

- Blue LED lights up if $T < 20^\circ\text{C}$;
- Red LED lights up for $T > 25^\circ\text{C}$.

Note that the two LEDs are connected antiparallel. The circuit should operate from a single $+V_s$ power supply. Plot the $V_{\text{out}}(I_T)$ characteristics of the circuit with the corresponding $V_{\text{out}}(T_{\text{°C}})$ curve.

8. Thermoelectric cooler control with NTC thermistor.

The resistance of the negative temperature coefficient (NTC) thermistors decreases with increasing temperature. Design a circuit using the **NTCLE100E3202** thermistor with nominal resistance of $R_{25} = 2 \text{ k}\Omega$ to measure the temperature of a sensing device and control the **MS2-065-04-04-11-11-11-W4** multistage thermoelectric (Peltier) cooler (TEC) to stabilise its operating temperature at $T_0 = 10^\circ\text{C}$. The TEC should:

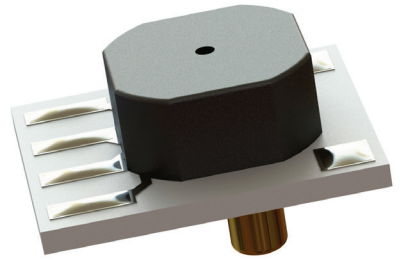


- Heat when $T < T_0 - 2^\circ\text{C}$;
- Cool when $T > T_0 + 2^\circ\text{C}$.
- Otherwise no current must flow.

Additionally, utilize small power blue, red and green LEDs of arbitrary types to indicate cooling (blue), heating (red), and T_0 is maintained (green). The hot side of the TEC is at room temperature of 20°C . From the TEC datasheet, select an appropriate operating current and voltage to remove $Q_c > 0.2 \text{ W}$ heat from the cold side with a coefficient of performance of $COP = \frac{Q_c}{P_{in}} > 1.5$. Specify the appropriate supply voltage. Plot the $I_{TEC}(R_{NTC})$ characteristics of the circuit with the corresponding $P_{in}(T)$ characteristics.

9. Pressure bridge sensor with dual colour LED and voltmeter readout.

The heart of the **RS12-0100** pressure sensor is a closed Wheatstone bridge comprised of four resistors on a silicon etched diaphragm. As pressure is applied to the diaphragm, the resistors are stressed, changing their resistance. If a constant voltage is applied to the bridge, the Δp pressure relative to the ambient pressure result in a voltage change ΔV at the bridge output. The maximum relative pressure is $\Delta p = 689 \text{ kPa}$. Design a circuit that measures Δp in the range of $-100 \dots + 400 \text{ kPa}$ and produces pressure values directly readable by a digital voltmeter as $-1 \dots + 4 \text{ V}$. Moreover, apply the **OSRPM25A32A** dual colour LED to indicate:



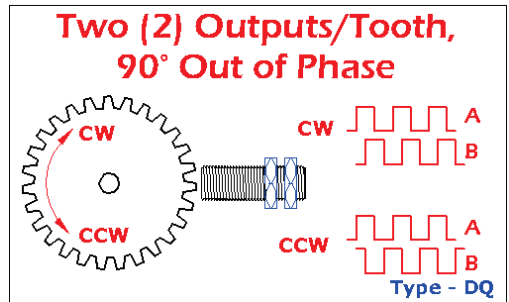
- If the pressure is 10% lower than ambient, green LED lights up for negative values;
- If the pressure is 10% higher than ambient, red LED lights up for positive values.

Note that the two LEDs are connected antiparallel. Specify the appropriate supply voltage. Plot the $V_{\text{out}}(\Delta V)$ characteristics of the circuit with the corresponding $V_{\text{out}}(\Delta p)$ curve.

10. Magnetic speed sensor with voltmeter readout.

A 2-channel, magnetic gear tooth sensor such as the **S56FW-37ADQ-5KCP4**, can display both speed and direction of a rotating gear when connected to a 2-channel oscilloscope. To display speed by a digital voltmeter as a DC voltage, design a circuit that:

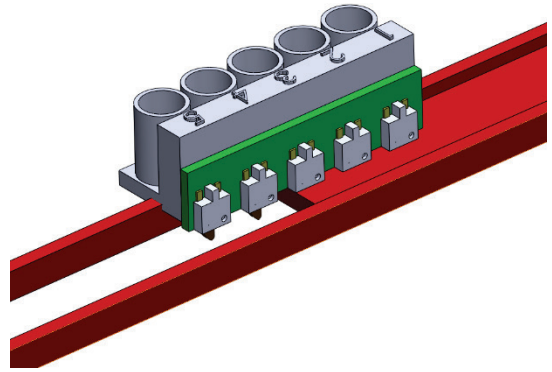
- Connects the sensor so that a symmetric $-5\text{ V} \dots +5\text{ V}$ square wave output is produced;
- Transforms the 50% duty cycle square wave output of one channel to a triangle wave;
- Detects the peak value of the triangle wave; and
- Converts the signal into the $+0.1\text{ V} \dots +10\text{ V}$ range that monotonically increases with the rotation frequency (speed) between $100\text{ Hz} \dots 1\text{ kHz}$. The increase may not be linear.
- Filters the signal before measuring to eliminate residual ripple.



By appropriately selecting the time constant of the conversion, the triangle wave will reach different peak values depending on the frequency of the signal within the detection range of the sensor. The next stage of the circuit records the peak value of the triangle signal that can be displayed by the **RS-12** digital voltmeter after proper low-pass filtering. Specify the appropriate supply voltage. Plot the $V_{\text{out}}(f)$ characteristics of the circuit.

11. Analog detection of discrete positions.

In some multi-tray measurements, it is important to register which of the sample trays are being tested. Tray positions can be detected using mechanical edge-detect switches (used also in media players to detect presence of e.g. disc).



Here a set of 5-position trays are fitted with 5 **MCTE-MRR-V** edge-detect switches as shown in the figure. The switches are normally open and one closes (shorting its two pins) when the supporting rail presses its button towards its body. Position- N is detected if the first N switches are closed by the rail ($N = 3$ in the figure).

First, tray position is to be indicated using the **LBG0500G3B** linear, 5-emitter LED-bargraph. Design your **analog** logic circuitry so that:

- Only a single LED, in position N , lights up when tray N is indicated.
- The circuit is able to drive each LED with the same current.

Second, the tray position N must also be output as **voltage proportional to N** (to provide possibility for recording by an analog-to-digital converter or digital voltmeter). Design this output with the necessary reference voltage keeping in mind that the circuit should operate from a single +5 V power supply. Plot the $V_O(N)$ characteristics of the circuit.

12. Analog gas flow meter.

The **SFM3020** is a gas flow meter for medical ventilators. Gas (air or O_2) flow in the sensor is unidirectional, yet allowing detection of back flow down to -10 slm (standard liter per minute). The analog output provides an analog linear voltage, representing the flow value in [slm] calibrated for air. The sensor is not fully bi-directional: output represents $-10 \dots +100\%$ full-scale.



First, gas flow rate (FR) is to be indicated using the **LBG1000T3-8G1Y1R** linear, 10-emitter LED-bargraph with custom colors: 1 red, then 1 yellow followed by 8 green emitters. Design your **analog** logic circuitry so that:

- Red LED lights up only when back flow is detected: $FR < -0.01$ slm.
- Yellow LED lights up only when no flow is detected (defined as flow is bound by output voltage noise): $|FR| < 0.01$ slm.
- Green LEDs light up if $FR > +0.01$ slm: the number of shining emitters is proportional to the flow rate, so that 1 ... 8 LEDs are equidistantly distributed in the entire forward flow range.

Second, the sensor flow rate voltages must also be converted to the $-0.1 \dots +1.6$ V **voltage range proportional to** [slm] showing sign of direction (to provide possibility for recording by an analog-to-digital converter or digital voltmeter). Design this output with the necessary reference voltage keeping in mind that the circuit should operate from a single $+5$ V power supply. Plot the $V_O(FR)$ characteristics of the circuit.